

Spazio Backend Setup & Development Guide

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For: Backend Developer

Project: Spazio (spaziocrypto.com)

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System Requirements

Software Dependencies

- Node.js: v22.22.1 (required - use nvm to manage versions)
- npm: v10.x or higher
- SQLite3: v3.x (included with most systems)
- Ghost-CLI: Latest stable
- Git: For version control

Hardware

- RAM: Minimum 2GB (4GB recommended)
- Disk: 5GB free space
- macOS, Linux, or Windows (WSL2)

Development Tools

- Code editor: VS Code, WebStorm, or similar
- Terminal/CLI: bash, zsh, or equivalent
- REST client: Postman or similar (for API testing)

Initial Setup

Step 1: Download Ghost Backend

Request the `ghost-local.zip` file from your frontend lead (Hamza).

```
# Extract the archive unzip ghost-local.zip cd ghost-local
```

Step 2: Install Node Version Manager (nvm)

Ghost v6 requires Node 22.

```
# macOS/Linux curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.39.0/install.sh  
| bash source ~/.bashrc # Verify installation nvm --version # Install Node 22 nvm  
install 22.22.1 nvm use 22.22.1 node --version # Should be v22.22.1
```

Step 3: Install Ghost-CLI

```
npm install -g ghost-cli # Verify ghost --version
```

Step 4: Configure Ghost Paths

Edit `config.development.json` to match your system:

```
# On macOS/Linux nano config.development.json
```

Update paths if needed:

```
{ "paths": { "contentPath": "/path/to/your/ghost-local/content" }, "database": {  
  "client": "sqlite3", "connection": { "filename":  
    "/path/to/your/ghost-local/content/data/ghost.db" } } }
```

Step 5: Install Dependencies

```
cd ghost-local/versions/6.21.0 npm install cd ../../
```

Backend Architecture

Directory Structure

```
ghost-local/ ■■■■ current # Symlink to active version ■■■■ versions/ ■ ■■■■ 6.21.0/ #
Ghost core application ■ ■■■■ core/ # Ghost internals ■ ■■■■ index.js # Entry point ■
■■■■ node_modules/ # Dependencies ■■■■ content/ # All user data ■ ■■■■ data/ ■ ■ ■■■■
ghost.db # SQLite database (posts, tags, members) ■ ■■■■ themes/ ■ ■ ■■■■
spazio-pitch-multi/ # Frontend theme ■ ■■■■ images/ # Uploaded images ■ ■■■■ files/ #
Uploaded documents ■ ■■■■ settings/ # Ghost configuration ■ ■■■■ logs/ # Application
logs ■■■■ config.development.json # Ghost configuration ■■■■ .ghostpid # Process ID file
```

Request Flow

```
Browser Request ↓ Ghost Server (Node.js) ↓ Theme Router (Handlebars templates) ↓
Content API / Admin API ↓ SQLite Database ↓ Response (HTML/JSON)
```

Running the Backend

Start Ghost

```
cd ghost-local nvm use 22.22.1 # Start the server ghost start # Or run interactively  
(helpful for debugging) ghost run # Check status ghost status
```

Access Ghost

- **Admin Panel**: <http://localhost:2368/admin>
- **Frontend**: <http://localhost:2368>
- **API URL**: <http://localhost:2368/ghost/api/v3/admin/>

Stop Ghost

```
cd ghost-local ghost stop ghost restart
```

API Integration Points

1. Content API (Public)

Used by theme to fetch posts, tags, members (read-only).

****Endpoint**:** `/ghost/api/v3/content/``

****Authentication**:** API Key only

****Key Endpoints**:**

```
// Get all posts GET /ghost/api/v3/content/posts/?key=YOUR_API_KEY // Get specific post
GET /ghost/api/v3/content/posts/{id}?key=YOUR_API_KEY // Get all tags GET
/ghost/api/v3/content/tags/?key=YOUR_API_KEY // Get members (requires member token) GET
/ghost/api/v3/content/members/?key=YOUR_API_KEY
```

****Example cURL**:**

```
curl "http://localhost:2368/ghost/api/v3/content/posts/?key=YOUR_API_KEY"
```

2. Admin API (Private)

Used for creating/editing posts, members, webhooks (requires authentication).

****Endpoint**:** `/ghost/api/v3/admin/``

****Authentication**:** JWT token

****Key Endpoints**:**

```
// Create post POST /ghost/api/v3/admin/posts/ // Update post PUT
/ghost/api/v3/admin/posts/{id}/ // Create member POST /ghost/api/v3/admin/members/ //
Delete post DELETE /ghost/api/v3/admin/posts/{id}/
```

3. Webhooks

Trigger actions when Ghost events occur (post published, member signup, etc.).

Setup:

```
# In Ghost Admin: Settings → Webhooks # Add webhook URL:  
https://your-backend/webhook/ghost/post-published
```

Webhook Events:

- `post.published`
- `post.unpublished`
- `post.deleted`
- `member.added`
- `member.deleted`
- `member.edited`

4. External APIs Already Configured

Formspree (Contact Form)

- **Endpoint:** `https://formspree.io/f/xwkdojlk`
- **Method:** POST contact form submissions
- **Frontend:** ``page-contact-us.hbs``

n8n Webhook (Records/Users)

- **Endpoint:** `https://n8n.ghostflow.net/webhook/2e340949-9ddd-4782-8a1e-6c1a54be6c16`
- **Purpose:** Fetch user records by ID
- **Frontend:** ``custom-post-records.hbs``

Cloudflare Worker (Multilingual Routing)

- **Endpoint:** `https://spazio-multilang.ghostflow.workers.dev`
- **Purpose:** Map posts to translated URLs
- **Frontend:** ``assets/js/languageLinks.js``

OneSignal (Push Notifications)

- **IDs Configured:** 9 language variants

- **Setup**: Settings → Integrations → OneSignal

Development Workflow

1. Local Development

```
# Terminal 1: Start Ghost cd ghost-local nvm use 22.22.1 ghost run # Terminal 2: Make
code changes cd ghost-local/versions/6.21.0/core/web/ # (Edit custom code, hooks,
middleware)
```

2. Custom Code Location

Add custom routes/middleware here:

```
ghost-local/versions/6.21.0/core/web/ ■■■ routes/ ■■■ middleware/ ■■■ services/ ■■■
api/
```

3. Create Custom API Route

Create file: `custom-api.js`

```
const express = require('express'); const router = express.Router(); // Custom
endpoint: GET /api/custom/market-data router.get('/market-data', (req, res) => { //
Fetch from CoinGecko or database res.json({ prices: [...] }); }); module.exports =
router;
```

Register in Ghost core:

```
// In versions/6.21.0/core/server/services/routing/index.js const customApi =
require('./custom-api'); app.use('/api/custom', customApi);
```

4. Testing API Locally

```
# Using curl curl http://localhost:2368/api/custom/market-data # Using Postman GET
http://localhost:2368/api/custom/market-data
```

Database & Content

SQLite Database

Location: `ghost-local/content/data/ghost.db`

Database Schema

Main tables:

- `posts` - Blog articles
- `tags` - Topic categories
- `post\_tags` - Post-tag relationships
- `members` - Subscriber accounts
- `members\_subscribe\_events` - Newsletter signups
- `users` - Admin users

Query Database

```
# Install sqlite3 CLI brew install sqlite3 # Query database sqlite3
ghost-local/content/data/ghost.db # Common queries SELECT * FROM posts LIMIT 10; SELECT
* FROM members; SELECT * FROM tags;
```

Backup Database

```
# Before making changes cp ghost-local/content/data/ghost.db ghost.db.backup # Restore
if needed cp ghost.db.backup ghost-local/content/data/ghost.db ghost restart
```

Custom Hooks & Extensions

Ghost Hooks System

Ghost provides lifecycle hooks for custom logic.

Register Custom Hook

```
// In custom module const ghostEvents = require('@tryghost/social-email/lib/events');
ghostEvents.on('post.published', async (post) => { console.log('New post published:',
post.title); // Custom logic here // - Notify external services // - Update database //
- Trigger webhooks });
```

Available Hooks

- `post.published` - When post goes live
- `post.deleted` - When post is removed
- `member.added` - New subscriber
- `member.deleted` - Subscriber removed
- `email.sent` - Newsletter sent
- `email.failed` - Newsletter error

API Integration Tasks

Task 1: Implement CoinGecko API

Goal: Fetch live crypto prices for market dashboard.

File: Create `services/crypto-service.js`

```
const axios = require('axios'); async function getCryptoPrices() { try { const response = await axios.get( 'https://api.coingecko.com/api/v3/simple/price', { params: { ids: 'bitcoin,ethereum', vs_currencies: 'usd', include_24hr_change: true } } ); return response.data; } catch (error) { console.error('CoinGecko error:', error); throw error; } } module.exports = { getCryptoPrices };
```

Expose via API:

```
router.get('/prices', async (req, res) => { const prices = await getCryptoPrices(); res.json(prices); });
```

Cache Strategy: Store in Redis or memory with 5-min TTL.

Task 2: Wire Newsletter Subscription

Goal: Save email signups to Ghost members.

File: Create `services/newsletter-service.js`

```
const ghostApi = require('@tryghost/api'); async function subscribeEmail(email, name = 'Subscriber') { try { const member = await ghostApi.members.create({ email: email, name: name, subscribed: true, labels: ['newsletter'] }); return member; } catch (error) { console.error('Subscription error:', error); throw error; } } module.exports = { subscribeEmail };
```

Task 3: Setup n8n Workflow

Goal: Fetch user records from external database.

****n8n Workflow Steps**:**

1. Webhook receives `GET /webhook/records?id=123`
2. Query database for record ID 123
3. Transform data to JSON
4. Return to frontend

****n8n Config**** (Already set up at: <https://n8n.ghostflow.net>)

Deployment

To Production Server

```
# 1. Build/test locally ghost stop ghost run # Test mode # 2. Upload to server rsync
-avz ghost-local/ user@server:/var/www/ghost-local/ # 3. On server cd
/var/www/ghost-local nvm use 22.22.1 ghost start # 4. Verify curl
https://yourdomain.com
```

Using Docker (Recommended)

Create `Dockerfile`:

```
FROM node:22.22.1 WORKDIR /ghost COPY ghost-local/ /ghost/ RUN npm install EXPOSE 2368
CMD ["ghost", "run"]
```

Build and run:

```
docker build -t spazio-ghost . docker run -p 2368:2368 spazio-ghost
```

CI/CD Pipeline

For GitHub Actions, create `.github/workflows/deploy.yml`:

```
name: Deploy Ghost on: push: branches: [main] jobs: deploy: runs-on: ubuntu-latest
steps: - uses: actions/checkout@v2 - uses: actions/setup-node@v2 with: node-version:
'22.22.1' - run: npm install - run: npm test - run: rsync -avz ./
user@server:/var/www/ghost-local/
```


Environment Variables

Create `.env` file in `ghost-local/`:

```
# Database GHOST_DB_PATH=/path/to/ghost-local/content/data/ghost.db # APIs
COINGECKO_API_KEY=your_key_here MAILCHIMP_API_KEY=your_key_here
N8N_WEBHOOK_URL=https://n8n.ghostflow.net/webhook/... # Email
MAIL_FROM=noreply@spaziocrypto.com SMTP_HOST=smtp.sendgrid.net SMTP_USER=apikey
SMTP_PASS=your_sendgrid_key # Ghost GHOST_URL=http://localhost:2368
GHOST_ADMIN_URL=http://localhost:2368/admin
```

Load in code:

```
require('dotenv').config(); const coingeckoKey = process.env.COINGECKO_API_KEY;
```

Troubleshooting

Ghost Won't Start

```
# Check Node version node --version # Should be v22.22.1 # Switch to correct version nvm
use 22.22.1 # Check process lsof -i :2368 # See if port 2368 is in use # Kill existing
process kill -9 <PID> # Start fresh ghost restart
```

Database Locked

```
# Stop Ghost ghost stop # Remove lock file rm ghost-local/.ghostpid # Start again ghost
start
```

API Returns 401 Unauthorized

- Verify API key in `config.development.json`
- Check JWT token expiration
- Regenerate API key in Ghost Admin

Theme Not Updating

```
# Clear cache rm -rf ghost-local/content/themes/spazio-pitch-multi/.cache # Restart
ghost restart
```

Memory Leak

```
# Monitor memory usage node --max-old-space-size=2048 ... # Or restart daily via cron 0
3 * * * cd /path/to/ghost-local && ghost restart
```

Quick Reference Commands

```
# Navigation cd ghost-local nvm use 22.22.1 # Start/stop ghost start ghost stop ghost
restart ghost run # Interactive mode # Status ghost status ghost log # View logs #
Database sqlite3 content/data/ghost.db # API Testing curl http://localhost:2368/api/...
# Deployment rsync -a ./ /destination/ ghost update # Update Ghost version
```

Support & Resources

- **Ghost Docs**: <https://ghost.org/docs/>
- **Ghost API**: <https://ghost.org/docs/api/>
- **Issue**: Slack/Email the frontend lead (Hamza)
- **Logs**: ``ghost-local/content/logs/``

Next Steps

1. Extract `ghost-local.zip`
2. Run setup steps 1-4 above
3. Start Ghost: `ghost start`
4. Access admin: <http://localhost:2368/admin>
5. Create test post to verify
6. Begin implementing APIs (CoinGecko, newsletter, etc.)
7. Push changes to GitHub
8. Deploy to staging server

****Last Updated****: April 1, 2026

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